



5.4 Kinematics using calculus

Calculus · IB Diploma · Concept guide and original exam-style practice

Understand the concept

For a particle with displacement $s(t)$, velocity is $v(t)=ds/dt$ and acceleration is $a(t)=dv/dt$; conversely, displacement can be recovered from velocity by integration, $s(t)=\int v(t)dt$, using an initial condition to determine the constant of integration. Displacement is the net change in position over an interval ($\int v(t)dt$, where positive and negative contributions cancel), whereas total distance travelled sums the absolute value of displacement over each sub-interval of constant velocity sign — direction reversals add rather than cancel.

Worked example: follow the reasoning

A particle moves with velocity $v(t)=3t^2-12t+9$ (m/s), $t \geq 0$. Find (a) the displacement between $t=0$ and $t=4$ s, (b) the times at which the particle is at rest, (c) the total distance travelled between $t=0$ and $t=4$ s.

Solution: (a) $\int_{[0,4]}(3t^2-12t+9)dt=[t^3-6t^2+9t]$ from 0 to 4 $= (64-96+36)=4$ m. (b) $v(t)=0$: $t^2-4t+3=0 \rightarrow t=1$ or $t=3$. (c) $s(t)=t^3-6t^2+9t$ gives $s(0)=0$, $s(1)=4$, $s(3)=0$, $s(4)=4$. Distances: $|4-0|+|0-4|+|4-0|=4+4+4=12$ m.

Quick check · multiple choice

A particle's velocity is $v(t)=2t-6$. At what time is the particle momentarily at rest?

- A $t=2$
- B $t=3$
- C $t=6$
- D $t=-3$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A particle moves along a straight line with velocity $v(t)=t^2-5t+6$ (m/s) for $0 \leq t \leq 5$, where t is in seconds. (a) Find the acceleration function $a(t)$. (b) Find the times when the particle is momentarily at rest. (c) Given the particle starts at $s(0)=0$, find the total distance travelled between $t=0$ and $t=5$, given the particle changes direction at each rest time found in (b). [8]

Final answer: _____



Answers and explanation

Question 1 [8]

(a) $a(t)=v'(t)=2t-5$. (b) $v(t)=0$: $t^2-5t+6=0 \rightarrow (t-2)(t-3)=0$, so $t=2$ and $t=3$. (c) $s(t)=t^3/3-2.5t^2+6t$: $s(0)=0$, $s(2)=14/3 \approx 4.667$, $s(3)=4.5$, $s(5) \approx 9.167$. Distances: $|4.667-0|+|4.5-4.667|+|9.167-4.5|=4.667+0.167+4.667=9.5$ m.

Quick check answer

B. $v(t)=0$ gives $2t-6=0$, so $t=3$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.